

Standard Industrial Dyeing Recipes & Process Manual

1. Exhaust Dyeing of 100% Cotton with Reactive Dyes (60°C)

Substrate: 100% Cotton Knit/Woven | **MLR:** 1:8 | **Target Shade:** Medium (2.0% owf)

Component / Auxiliary	Dosage	Function
Reactive Dye (Red 195)	2.0% owf	Primary Colorant
Glauber's Salt (Na2SO4)	45 g/L	Electrolyte for dye exhaustion
Soda Ash (Na2CO3)	15 g/L	Alkali for covalent fixation
Sequestering Agent	1.0 g/L	Chelates Ca2+/Mg2+ ions
Levelling Agent	1.0 g/L	Ensures uniform wetting

2. HTHP Exhaust Dyeing of 100% Polyester with Disperse Dyes (130°C)

Substrate: 100% Polyester | **MLR:** 1:6 | **Target Shade:** Dark Navy/Black (4.0% owf)

Component / Auxiliary	Dosage	Function
Disperse Dyes (HT/SE type)	4.0% owf	High thermal fastness colorant
Dispersing Agent	1.5 g/L	Prevents dye agglomeration
HT Leveling Agent	1.0 g/L	Promotes dye migration at 110-130°C
Acetic Acid (80%)	0.8 - 1.2 g/L	Adjusts pH to 4.5 - 5.5
Reduction Clearing (RC)	2 g/L Hydros + 2 g/L Causin	Removes surface dye at 80°C for 20 min

3. Acid Dyeing of Polyamide (Nylon 6 / Nylon 6,6)

Substrate: Nylon Knits | **MLR:** 1:10 | **Temp:** 98°C - 100°C | **pH:** 4.8 - 5.2

Recipe: Acid Dye 1.5% owf, Ammonium Sulfate 1-2 g/L, Anionic Leveler 1.0 g/L. Post-treatment: Syntan Fixing Agent 2.5% owf at 70°C for 20 min.